Nonlinear dependence without low-dimensional chaos:

a surrogate-based study of returns and volatility in ten cryptocurrencies and the Nifty 50 constituents

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Abstract

Whether financial prices are generated by low-dimensional deterministic chaos has been asked repeatedly since the late 1980s, and the answer has depended heavily on the null hypothesis against which chaos diagnostics are judged. We revisit the question with a uniform, pre-specified protocol applied to 61 daily series: the ten largest cryptocurrencies, the fifty current constituents of the Nifty 50 index and the index itself. For each asset we study log returns, absolute returns, squared returns and 20-day realised volatility, and compute delay-embedding parameters, the largest Lyapunov exponent, the correlation dimension, the 0–1 test for chaos, nonlinear prediction skill, S-map nonlinearity gain and time-reversal asymmetry. Every statistic is compared with 999 iterative amplitude-adjusted Fourier transform (IAAFT) surrogates and 999 surrogates simulated from a fitted GARCH(1,1) model with Student-t innovations, with false discovery rate control within and across assets. We complement the time-domain analysis with Welch power spectra, spectral exponents and band shares, tested against the same surrogate families over the full sample and four market periods. The conventional chaos indicators are uninformative on their own: the largest Lyapunov exponent is positive in 243 of 244 series, the correlation dimension appears to saturate in most series and the 0–1 statistic is close to one everywhere, yet GARCH surrogates reproduce all three. Nonlinear structure relative to a linear Gaussian null is widespread (all ten cryptocurrencies and 42 of 51 Nifty series in returns), but after cross-asset correction only 2 of 1,464 tests against GARCH remain significant, both isolated correlation-dimension results. No series is classified as potentially chaotic. The one robust feature that GARCH does not reproduce is a seven-day spectral peak in the absolute returns of every cryptocurrency (and in realised volatility of five), which we interpret as a weekly calendar effect rather than as deterministic dynamics. The spectral shape of volatility also differs systematically between markets: relative to GARCH, cryptocurrency volatility is less persistent, and squared returns of about a third of the stocks carry more low-frequency power.

Keywords: deterministic chaos; surrogate data; nonlinearity; GARCH; power spectrum; cryptocurrencies; Nifty 50; multiple testing

1 Introduction

The possibility that the irregular behaviour of asset prices reflects low-dimensional deterministic dynamics, rather than the accumulation of exogenous random shocks, has attracted economists

and physicists for more than three decades. The appeal is easy to understand. A deterministic system with a small number of effective degrees of freedom would be predictable over short horizons even if it looked random to linear methods, and theoretical asset-pricing models with heterogeneous, boundedly rational traders can produce exactly such dynamics (Brock and Hommes, 1998). The empirical literature that followed, however, taught a sobering lesson: the standard diagnostics of chaos, namely a positive largest Lyapunov exponent, a low and saturating correlation dimension and apparent structure in reconstructed phase space, can all be produced by stochastic processes, particularly by processes with heavy tails, strong persistence or time-varying volatility (Osborne and Provenzale, 1989; Ruelle, 1990; Eckmann and Ruelle, 1992). When the same diagnostics are applied to residuals of models that capture conditional heteroskedasticity, most evidence for chaos in stock, currency and commodity returns disappears (Hsieh, 1989, 1991; Brock et al., 1991).

Two developments make the question worth revisiting. The first is methodological. Surrogate data testing (Theiler et al., 1992; Schreiber and Schmitz, 1996, 2000; Lancaster et al., 2018) and the 0–1 test for chaos (Gottwald and Melbourne, 2004, 2009) are now mature, and modern computing makes it possible to use several hundred surrogates per test, for many statistics, many series and many assets, which in turn makes proper control of the false discovery rate both possible and necessary (Benjamini and Hochberg, 1995; Harvey et al., 2016). The second is the emergence of cryptocurrency markets, which trade continuously, are dominated by retail and algorithmic participants, and display volatility and tail behaviour far more extreme than those of equity markets (Corbet et al., 2019). Several studies report chaos or strong nonlinear structure in Bitcoin (Lahmiri and Bekiros, 2018), but such conclusions typically rest on comparisons with linear or independent null models, which are known to be rejected by any series with volatility clustering.

This paper asks a deliberately narrow question: once the stylised facts of returns, namely heavy tails and volatility clustering, are respected by the null model, is there evidence of low-dimensional deterministic dynamics, or of any nonlinear structure beyond a standard volatility model, in daily returns and volatility of cryptocurrencies and of large Indian stocks? We make three contributions.

1. We apply a single, pre-specified protocol to 61 assets from two very different markets and four transformations of each return series, so that results are comparable across assets and are not the product of asset-by-asset tuning. The protocol combines geometric diagnostics (embedding, Lyapunov exponent, correlation dimension), a dynamical diagnostic (the 0–1 test), predictive diagnostics (simplex and S-map) and a symmetry diagnostic (time reversal).

2. Every statistic is judged against two surrogate families with different null hypotheses: IAAFT surrogates, which preserve the linear autocorrelation and the marginal distribution, and GARCH(1,1)-t surrogates, which additionally reproduce volatility clustering and heavy tails. Results are corrected for multiple testing both within each asset and across all assets, and checked for stability in the two halves of each sample.

3. We add a frequency-domain analysis that is usually absent from chaos studies in finance:

Welch power spectra, spectral exponents and band shares, compared with surrogate spectra over the full sample and over four market periods. This turns out to be where the only robust departure from the GARCH null appears.

The results are negative regarding chaos and informative regarding everything else. The classical diagnostics look "chaotic" in almost every series, but GARCH surrogates look equally chaotic. Nonlinearity relative to a linear null is pervasive, but it is almost entirely accounted for by conditional heteroskedasticity. No series is classified as potentially chaotic. What survives is a weekly cycle in cryptocurrency volatility and systematic differences between the two markets in the spectral shape of volatility, both of which point to calendar and persistence effects that a GARCH(1,1) model omits rather than to deterministic dynamics.

Section 2 reviews the literature. Section 3 describes the data, Section 4 the methods and Section 5 the results. Section 6 discusses the interpretation, and Section 7 concludes.

2 Literature review

2.1 From strange attractors to time-series diagnostics

The empirical study of chaos in observed data rests on the embedding theorem of Takens (1981), generalised by Sauer et al. (1991): for a deterministic system whose trajectories lie on an attractor of dimension D, a vector of m delayed observations $(x_t, x_{t-\tau}, \ldots, x_{t-(m-1)\tau})$ with $m > 2D$ generically reconstructs the attractor up to a smooth change of coordinates. Practical reconstruction requires choosing the delay τ and the dimension m. The delay is commonly taken at the first minimum of the average mutual information (Fraser and Swinney, 1986) or where the autocorrelation function has decayed sufficiently, and the dimension is the smallest m at which the fraction of false nearest neighbours vanishes (Kennel et al., 1992). Invariants of the reconstructed attractor then characterise the dynamics: the correlation dimension of Grassberger and Procaccia (1983), estimated from the scaling of the correlation sum, and the largest Lyapunov exponent, estimated from the average exponential divergence of initially close trajectories (Wolf et al., 1985; Rosenstein et al., 1993). Comprehensive treatments are given by Eckmann and Ruelle (1985) and Kantz and Schreiber (2004). Recurrence plots (Eckmann et al., 1987) and recurrence quantification (Marwan et al., 2007) provide visual and quantitative summaries of recurrences in phase space.

A different route to detecting chaos avoids phase-space reconstruction altogether. Gottwald and Melbourne (2004) proposed the 0–1 test, which drives a two-dimensional system with the observed series and measures whether the resulting trajectory is bounded (regular dynamics, $K \approx 0$) or diffusive (chaotic dynamics, $K \approx 1$); implementation issues such as oversampling and the choice of frequency are discussed in Gottwald and Melbourne (2009). A third route uses prediction. Sugihara and May (1990) showed that for chaotic systems the skill of nearest-neighbour (simplex) forecasts declines with horizon, whereas for uncorrelated noise it is flat, and Sugihara (1994) introduced the S-map, a locally weighted linear forecast whose improvement over a global linear model as the locality parameter increases measures state-dependent, that is nonlinear, dynamics.

2.2 The critique: stochastic processes that look chaotic

The early enthusiasm for these diagnostics was tempered by a series of results showing how easily they are misled. Theiler (1986) showed that temporal correlation produces spurious low correlation dimensions unless temporally close points are excluded. Osborne and Provenzale (1989) showed that coloured stochastic processes with power-law spectra have finite, saturating correlation dimensions. Ruelle (1990) and Eckmann and Ruelle (1992) derived the data requirements for meaningful dimension and Lyapunov estimates, which grow exponentially with the dimension and are rarely met by economic time series. The Rosenstein divergence curve has a positive initial slope for many noisy stochastic series, so a positive estimated exponent is a necessary but not a sufficient sign of chaos; and the 0–1 test returns $K \approx 1$ for uncorrelated noise as well as for chaos, so it cannot by itself distinguish the two (Gottwald and Melbourne, 2009).

The constructive response to these problems was the method of surrogate data (Theiler et al., 1992). A null hypothesis is specified as a class of processes, an ensemble of artificial series consistent with the null is generated from the data, and the discriminating statistic computed on the data is ranked within its distribution over the surrogates. Amplitude-adjusted Fourier transform surrogates test the null of a linear Gaussian process observed through a static monotone transformation; the iterative refinement of Schreiber and Schmitz (1996) matches the power spectrum and the marginal distribution more accurately. Schreiber and Schmitz (2000) and Lancaster et al. (2018) review surrogate methods and warn that rejection of a linear null says only that the series is not linear Gaussian; it does not say what it is. In financial data, where volatility clustering guarantees rejection of linear nulls, model-based surrogates that preserve conditional heteroskedasticity are the natural next step. Time-reversal asymmetry provides a complementary, model-free signature: linear Gaussian processes and their static transforms are statistically reversible, so asymmetry indicates nonlinearity or non-Gaussian driving (Diks et al., 1995; Ramsey and Rothman, 1996).

2.3 Chaos and nonlinearity in financial markets

Applications to financial data began in the late 1980s. Scheinkman and LeBaron (1989) found that the correlation dimension of weekly US stock returns was lower than that of shuffled returns, and proposed the comparison with shuffled data that underlies the BDS test (Brock et al., 1996). Frank and Stengos (1989) reported low dimensions for gold and silver returns, and Peters (1991) popularised chaos and fractal ideas in the practitioner literature. The decisive counter-evidence came from studies that applied the same tests to standardised residuals of GARCH-type models: Hsieh (1989) for exchange rates and Hsieh (1991) for stock returns found that nonlinear dependence is largely explained by conditional heteroskedasticity and found no convincing evidence of chaos, a conclusion summarised in the monograph of Brock et al. (1991). Mayfield and Mizrach (1992) estimated dimensions for high-frequency S&P 500 data and found them high, and Abhyankar et al. (1997) found nonlinear dependence but no evidence of low-dimensional chaos in four major stock indexes at high frequency. Barnett and Serletis (2000) reviewed the econometric evidence and concluded that tests for chaos in economic data are inconclusive at best and that their power against realistic stochastic alternatives is poorly understood; Kyrtsou and

Serletis (2006) reached similar conclusions with a battery of univariate tests. On the linearity side, tests such as those of Hinich (1982) and Tsay (1986) routinely reject linearity in returns, consistent with the stylised facts collected by Cont (2001): heavy tails (Mandelbrot, 1963), volatility clustering (Engle, 1982; Bollerslev, 1986, 1987), and slow decay of the autocorrelation of absolute returns (Ding et al., 1993).

2.4 Volatility persistence, long memory and the spectrum

Spectral methods have a long history in economics; Granger (1966) documented the "typical spectral shape" of economic variables, with power concentrated at low frequencies. For volatility, the slow, hyperbolic decay of autocorrelations in absolute and squared returns (Ding et al., 1993) corresponds to a spectrum that diverges as a power law at low frequency, $S(f) \propto f^{-\beta}$ with $\beta = 2d$ for a fractionally integrated process of order d. The memory parameter can be estimated semiparametrically from the low-frequency periodogram (Geweke and Porter-Hudak, 1983), and long-memory volatility models such as FIGARCH (Baillie et al., 1996) and the heterogeneous autoregressive model of realised volatility (Corsi, 2009; Andersen et al., 2003) were proposed to capture it. At the same time, apparent long memory can be generated by structural breaks or slowly changing parameters in short-memory models: Lamoureux and Lastrapes (1990) showed that breaks inflate the estimated persistence of GARCH models, and Mikosch and Stărică (2004) showed that nonstationarity can produce both apparent long-range dependence and integrated GARCH estimates. Lo (1991) showed that classical rescaled-range tests confuse short-range dependence with long memory. Periodic components are also well known in volatility: Andersen and Bollerslev (1997) showed that intraday periodicity in volatility distorts estimates of volatility persistence if it is not removed. These results are directly relevant to chaos testing, because a power-law spectrum or a persistent volatility process is exactly the kind of stochastic structure that misleads dimension and Lyapunov estimates (Osborne and Provenzale, 1989). In the physics literature, offset log-scale spectra across regimes are a standard way to compare dynamics between parameter ranges (e.g. Kailasham and Khair, 2023); we borrow this presentation for comparing market periods.

2.5 Cryptocurrencies

The cryptocurrency literature has focused on market efficiency, volatility modelling and calendar effects. Urquhart (2016) found that Bitcoin returns were inefficient in its early years and becoming more efficient over time, and Bariviera (2017) reached similar conclusions with time-varying Hurst exponents. Katsiampa (2017) compared GARCH specifications for Bitcoin and found that models with a long-run and a short-run volatility component fit best. Lahmiri and Bekiros (2018) applied the largest Lyapunov exponent and entropy measures to Bitcoin prices and reported chaos and multifractality, without model-based surrogates. Corbet et al. (2019) survey the field. On calendar effects, the evidence is mixed for returns but clearer for activity and volatility: Baur et al. (2019) found weekday and weekend differences in Bitcoin trading volume and volatility, Caporale and Plastun (2019) found a day-of-the-week effect in Bitcoin returns but not in other coins, and Aharon and Qadan (2019) reported day-of-the-week effects in both Bitcoin returns and volatility. In equity markets the weekend effect in returns is a classic

anomaly (French, 1980). To our knowledge, no previous study has linked the weekly calendar structure of cryptocurrency volatility to the question of nonlinear or chaotic dynamics, or shown that it is the main feature that a GARCH null fails to reproduce.

2.6 Gap addressed by this paper

Three gaps motivate the present study. First, most chaos studies of individual assets use a single null model, often linear, so that rejection is guaranteed by volatility clustering and is easily over-interpreted. Second, studies rarely apply the same protocol to many assets with control of the false discovery rate across assets, although with dozens of statistics per series the expected number of false positives is substantial (Benjamini and Hochberg, 1995; Harvey et al., 2016). Third, frequency-domain evidence and time-domain chaos diagnostics are rarely examined together, although the former explains many of the artefacts of the latter. Indian equities, despite the size of the market, have received comparatively little attention in this literature, and a direct comparison of a 24/7 cryptocurrency market with an exchange-traded equity market under one protocol is, as far as we know, new.

3 Data

Cryptocurrencies. We use daily closing prices (00:00 UTC) of the ten cryptocurrencies with the largest trading volume at the time of download, quoted against the US dollar stablecoin USDT on a large spot exchange: BTC, ETH, BNB, SOL, XRP, DOGE, ADA, AVAX, TRX and DOT. Each series starts on the first available trading day (2017-09-06 for BTC and ETH; between 2017-11 and 2020-10 for the others) and ends on 2026-09-02. The number of daily returns ranges from 2,152 (AVAX) to 3,284 (BTC, ETH), with a median of 3,005. Cryptocurrencies trade seven days a week, so a "day" is a calendar day.

Nifty 50. We use daily dividend- and split-adjusted closing prices of the fifty stocks in the Nifty 50 index at the time of download, and the closing level of the index itself, from 2016-01-01 to 2026-10-16. The price panel was cleaned before this study: days on which nearly all stocks reported zero volume and unchanged prices were removed, and two corporate actions not reflected in the source data were corrected. Most stocks have 2,644 daily returns. Six were listed after 2016; the shortest histories are JIOFIN (758 returns) and ETERNAL (1,273 returns), which we flag as "low power" (between 750 and 1,499 observations); every other asset has at least 1,500 observations. The constituent list is the current one, so the sample is subject to survivorship bias. The data were read only and never modified.

Series analysed. For each asset we compute log returns $r_t = \ln P_t - \ln P_{t-1}$ and four series: returns r_t, absolute returns $|r_t|$, squared returns r_t^2, and 20-day realised volatility $\mathrm{RV}_t = \left(\frac{1}{20}\sum_{j=0}^{19} r_{t-j}^2\right)^{1/2}$. The first 19 observations are dropped from all four series so that they are aligned. Daily return volatility is very different in the two markets: the median (across assets) standard deviation of daily returns is 0.051 for cryptocurrencies and 0.018 for Nifty stocks.

Market periods. For the period analysis we split each sample into four periods chosen before the analysis to separate broad market regimes: P1 (2017-09 to 2019 for crypto, 2016–2019 for Nifty), P2 (2020–2021, including the COVID-19 crash and the 2021 boom), P3 (2022–2023) and P4 (2024 to the end of the sample). A period is analysed for an asset only if the asset has at least 365 observations in it and at least half of the nominal period; this gives 6, 10, 10 and 10 cryptocurrencies and 48, 48, 50 and 51 Nifty series in P1 to P4.

4 Methods

All settings below were fixed before the cross-asset run and are identical for every asset. The pseudo-random generator is seeded deterministically for every surrogate, so all results are exactly reproducible.

4.1 Embedding

Each series is standardised. The delay τ is the smaller of the first local minimum of the average mutual information (16 equiprobable bins, lags up to 30) and the first lag at which the absolute autocorrelation falls below $1/e$. The embedding dimension m is the smallest $m \leq 10$ at which the fraction of false nearest neighbours (Kennel et al., 1992) falls below 5%; if no such m exists, $m = 10$ is used and the series is recorded as not having reached a low-dimensional embedding. Temporally close neighbours are excluded with a Theiler window of $\max(10, 2\tau)$ observations, and at least 20 for realised volatility, whose overlapping construction induces 20 days of mechanical dependence.

4.2 Chaos diagnostics

- *Largest Lyapunov exponent* (LLE): the algorithm of Rosenstein et al. (1993), with the slope of the mean log-divergence curve fitted from step 1 to the first step at which the curve has completed 60% of its total rise (at least four steps). The goodness of the linear fit is recorded.

- *Correlation dimension* D_2: correlation sums with the maximum norm on up to 2,000 embedded points (Grassberger and Procaccia, 1983), local slopes in the scaling range $10^{-3} \leq C(r) \leq 10^{-1}$, for $m = 1, \ldots, 10$. We say that D_2 saturates if some three consecutive values of m have slopes within 15% of their mean and that mean is below $m - 1$. This rule was validated on the logistic map before use. For surrogate testing, the statistic is the slope at $m = 5$.

- *0–1 test*: the correlation method of Gottwald and Melbourne (2009) applied to the series subsampled every τ steps, with the median K over 50 random frequencies $c \in (\pi/5, 4\pi/5)$.

- *Recurrence determinism* (DET): the share of recurrence points on diagonal lines of length at least 2, for the last 1,000 embedded points at a 10% recurrence rate. DET is reported descriptively only.

4.3 Nonlinearity diagnostics

- *Prediction skill*: the out-of-sample correlation between one-step simplex forecasts (Sugihara and May, 1990) and outcomes, with the first 70% of the sample as library.
- *S-map gain*: $1 - \mathrm{MSE}_{\theta^*}/\mathrm{MSE}_0$, where MSE_0 is the one-step error of a global linear autoregression on the embedding vector and MSE_{θ^*} is the smallest error of the S-map (Sugihara, 1994) for $\theta \in \{1, 2, 4\}$, both evaluated on the last part of the sample.
- *Time-reversal asymmetry*: $\langle (x_{t+1} - x_t)^3 \rangle / \langle (x_{t+1} - x_t)^2 \rangle^{3/2}$, which is zero in expectation for reversible processes (Diks et al., 1995; Ramsey and Rothman, 1996).

4.4 Surrogates and hypothesis tests

Two surrogate families are used, each with 999 members per series.

1. *IAAFT* (Schreiber and Schmitz, 1996): 200 iterations, generated directly from each of the four series. The null hypothesis is a stationary linear Gaussian process observed through a static monotone transformation.
2. *GARCH(1,1)-t*: a GARCH(1,1) model with constant mean and Student-t innovations (Bollerslev, 1986, 1987) is fitted by maximum likelihood to the returns of each asset; return paths of the same length are simulated (500 burn-in steps) and transformed into absolute returns, squared returns and realised volatility exactly as the data are. The null hypothesis is that the series is generated by this volatility model. It preserves heavy tails and volatility clustering but contains no deterministic dynamics.

For each statistic the rank p-value is $(1 + \#\{\text{surrogates at least as extreme}\})/(N + 1)$, one-sided where the direction is implied by the alternative (lower D_2, higher prediction skill, higher S-map gain) and two-sided otherwise. With $N = 999$ the smallest attainable p-value is 0.001 (one-sided) and 0.002 (two-sided). As an effect size we report z, the difference between the observed statistic and the surrogate mean divided by the surrogate standard deviation.

Multiple testing. Each asset has $4 \times 6 \times 2 = 48$ surrogate tests. Within an asset they are adjusted with the Benjamini–Hochberg procedure (Benjamini and Hochberg, 1995). Across assets, the 61 p-values of each test family (series $\times$ statistic $\times$ surrogate family) are adjusted together, and we call a result significant across assets if its cross-asset q-value is below 0.05.

Split-sample stability. The full set of tests is repeated on the first and second halves of each sample with 499 surrogates per family and independent seeds.

Classification rule. Each series is assigned to one class by a rule fixed in advance: *Periodic* or *Quasi-periodic* if $K < 0.2$ and D_2 saturates (distinguished by the value of D_2); *Potentially Chaotic* if $K > 0.8$, D_2 saturates and at least two of the deterministic statistics (D_2, prediction skill, S-map gain, LLE) are significant against *both* surrogate families; *Random/Stochastic* if $K > 0.8$ and no statistic is significant against GARCH; *Inconclusive* otherwise. A positive LLE, a saturating D_2 or a visual pattern alone never leads to a chaotic classification.

4.5 Power spectra

Spectra are estimated with Welch's method (Welch, 1967) on the standardised series (Hann window, 512-day segments, 50% overlap, linear detrending), giving a frequency resolution of 1/512 cycles per day. We report: (i) the spectral exponent β from an ordinary least squares fit of $\log S$ on $\log f$ over periods of 25 to 250 days, with its R^2; (ii) the semiparametric estimate $\beta_{\mathrm{GPH}} = 2\hat{d}$ of Geweke and Porter-Hudak (1983) using the lowest $\sqrt{n}$ Fourier frequencies; (iii) the shares of spectral power at periods longer than 20 days (low band) and shorter than 5 days (high band); (iv) the three largest spectral peaks with periods up to 250 days. Each quantity is compared with its distribution over 999 surrogate spectra of each family. The whole spectrum is tested with a global maximum-deviation statistic, $D = \max_f |\log S(f) - \mathrm{med}_0(f)|/\mathrm{sd}_0(f)$, computed identically for the data and each surrogate, which controls the family-wise error over frequencies. A weekly component is tested one-sided at the Welch frequency closest to one cycle per seven days for cryptocurrencies and one cycle per five trading days for stocks. This frequency was specified in advance as a calendar check; the cross-asset analysis is therefore a replication of the Bitcoin result, which was obtained first. Spectral p-values are adjusted within each asset and across assets in the same way as the surrogate tests.

Two features of the spectra are mechanical and are taken into account in interpretation. The 20-day moving average in realised volatility has spectral zeros at $f = k/20$ cycles per day, which appear as evenly spaced dips in its spectrum; the GARCH surrogates pass through the same filter, so comparisons with them remain valid. One of these zeros falls at $f = 0.2$, the five-day frequency, so the weekly test for realised volatility of stocks is uninformative by construction.

4.6 Period analysis

For each market period we compute time-domain summaries (mean, standard deviation, kurtosis, lag-1 autocorrelation) and Welch spectra with 256-day segments, the spectral exponent over periods of 25 to 128 days and the band shares. Each period is compared with 499 GARCH surrogates of a model refitted to that period only, and with 499 IAAFT surrogates, using the global spectral test and tests on β and the low-band share. Spectra of different periods are displayed with vertical offsets on a logarithmic axis, following the presentation of Kailasham and Khair (2023).

4.7 Computation

The full protocol requires about 40,000 surrogate series per asset. It was run on a shared workstation using at most half of its processors, with each analysis stage checkpointed to disk. The Bitcoin results obtained on this machine were checked against an independent run of the same code on a different computer and agree to the last reported digit.

Table 1: Embedding and chaos diagnostics across assets (medians across assets unless stated). FNN: share of assets for which the false-nearest-neighbour fraction reached 5% at some $m \leq 10$. LLE > 0 and D_2 sat.: share of assets with a positive Lyapunov exponent and a saturating correlation dimension.

Market	Series	τ	FNN	LLE	LLE > 0	D_2 sat.	D_2	K
Crypto (10)	returns	1	0.00	0.114	1.00	1.00	4.51	0.997
	absolute returns	1	0.00	0.138	1.00	1.00	4.00	0.998
	squared returns	1	0.30	0.203	1.00	0.80	2.52	0.999
	realised vol (20d)	22	0.00	0.066	1.00	0.90	5.22	0.994
Nifty (51)	returns	1	0.00	0.113	1.00	0.94	5.49	0.998
	absolute returns	1	0.00	0.133	1.00	0.94	4.98	0.998
	squared returns	1	0.10	0.189	0.98	0.98	2.98	0.998
	realised vol (20d)	21	0.00	0.066	1.00	0.78	5.42	0.996

5 Results

5.1 Embedding and conventional chaos indicators

Table 1 summarises the diagnostics across assets. Three observations stand out. First, no low-dimensional embedding is found: for returns, absolute returns and realised volatility, the false-nearest-neighbour fraction never falls below 5% for $m \leq 10$ in any of the 61 assets, and for squared returns it does so in only 3 cryptocurrencies and 5 Nifty series. The delay is one day for the return-based series and about 21 to 22 days for realised volatility, where it reflects the averaging window. Second, the conventional indicators would, if taken at face value, suggest chaos almost everywhere: the LLE is positive in 243 of 244 series, with median values between 0.07 and 0.20 per day; D_2 satisfies the saturation rule in 78% to 100% of series depending on the transformation, at values between 2.5 and 5.5; and the 0–1 statistic K lies between 0.97 and 1.00 in every series. Third, and decisively, GARCH surrogates produce the same values. For returns, the median LLE is 0.114 in the cryptocurrency data and 0.113 in their GARCH surrogates, and 0.113 against 0.113 for Nifty; the median D_2 at $m = 5$ is 4.00 against 4.12 (crypto) and 4.26 against 4.33 (Nifty); and the median K is 0.998 in both data and surrogates. A process with no deterministic component at all thus reproduces every conventional "signature of chaos" in these data.

5.2 Nonlinearity relative to a linear null

Against IAAFT surrogates, nonlinear structure is common (Figure 1, light bars). After cross-asset correction, returns differ from their IAAFT surrogates in all ten cryptocurrencies and in 42 of 51 Nifty series; absolute returns in 7 of 10 and 11 of 51; realised volatility in 6 of 10 and 5 of 51; squared returns in 1 of 10 and 5 of 51. In total 119 of 1,464 IAAFT tests are significant across assets. The statistic responsible is most often the LLE of returns (all ten cryptocurrencies and 41 Nifty series), followed by the S-map gain of returns (3 and 13) and time-reversal asymmetry in absolute returns and realised volatility. The sign of the LLE effect is informative: the exponent of the data is *smaller* than that of the IAAFT surrogates (median $z = -2.9$ for crypto, -2.8 for Nifty), the opposite of what an additional chaotic component would produce. Time-reversal asymmetry of volatility has opposite signs in the two markets:

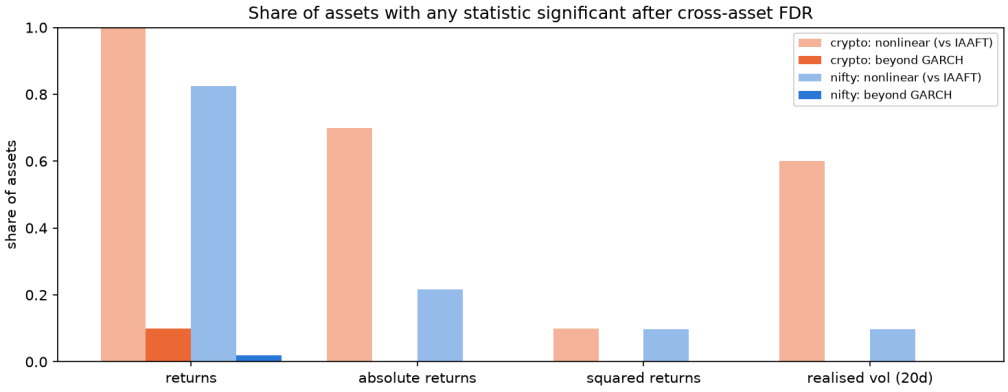


Figure 1: Share of assets in which at least one statistic is significant after cross-asset false discovery rate control, by series and market. Light bars: against IAAFT surrogates (nonlinearity relative to a linear Gaussian null). Dark bars: against GARCH(1,1)-t surrogates.

positive for every significant cryptocurrency series (median $z = 4.3$ for absolute returns and 10.4 for realised volatility) and negative for most significant Nifty series.

5.3 Nonlinearity relative to the GARCH null

Against GARCH surrogates the picture changes completely (Figure 1, dark bars, and Figure 2). Of the 1,464 tests against GARCH, two remain significant after cross-asset correction: the correlation dimension of DOGE returns ($z = -4.5$, $q = 0.03$) and of SBILIFE returns ($z = -3.1$, $q = 0.03$). Neither is accompanied by any other significant statistic in the same series, and neither replicates in the second half of its sample. At the nominal 5% level, 133 tests have $p \leq 0.05$ against an expectation of 73 under the global null, so there is a modest excess of small p-values, spread thinly across statistics and assets; the largest shares of $|z| > 1.96$ are in the S-map gain of absolute returns (20% of assets), the LLE of realised volatility (18%) and the time-reversal asymmetry of squared returns (18%). This pattern is what one would expect from a model that is slightly misspecified in many small ways (for example in its tail shape, its asymmetry, or its persistence) rather than from a strong common deterministic signal.

Within assets, 13 of the 244 series have at least one statistic significant against GARCH after within-asset correction (five cryptocurrency series and eight Nifty series), and all 13 are classified *Inconclusive* because none meets the conditions for a chaotic classification: in each case fewer than two deterministic statistics are significant against both surrogate families, or D_2 does not saturate. Of these 13, three (BSE, DOGE and HDFCLIFE) show the same departure in the first half of the sample and none in the second half. The final classification is 231 series *Random/Stochastic*, 13 *Inconclusive*, and none *Periodic*, *Quasi-periodic* or *Potentially Chaotic*.

5.4 Power spectra of returns and volatility

Return spectra are flat. The median spectral exponent of returns is 0.11 for cryptocurrencies and 0.03 for Nifty series, with R^2 of 0.03 and 0.08, and the share of power at periods above 20 days, 0.09 in both markets, equals the share expected for white noise. No whole-spectrum test against GARCH is significant across assets for any series in either market, and no return series

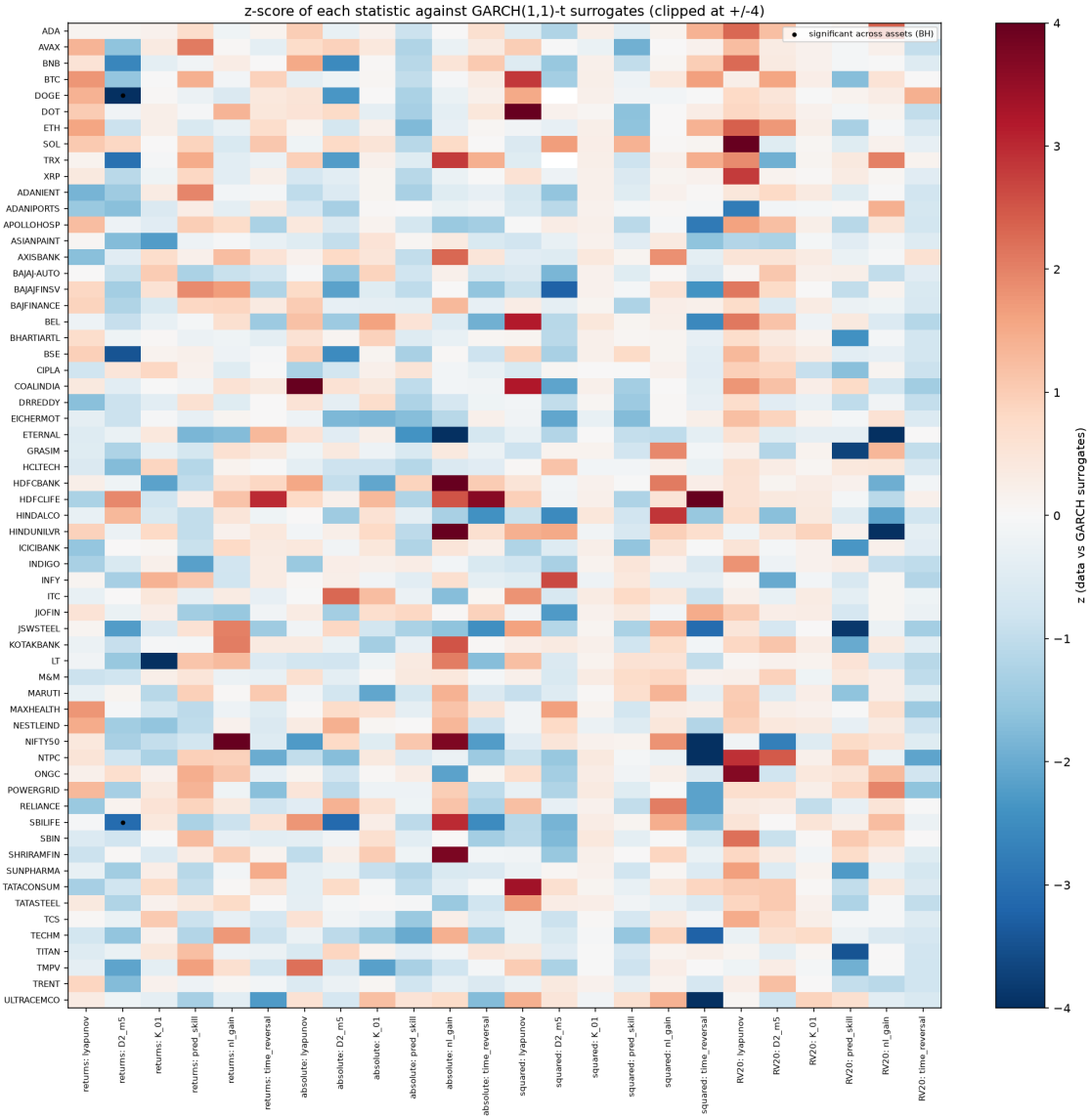


Figure 2: Effect size z of each statistic against GARCH(1,1)-t surrogates for every asset (rows) and every series and statistic (columns), clipped at ± 4. Black dots mark the two tests that are significant after cross-asset false discovery rate control.

Table 2: Spectral summaries (medians across assets). β: spectral exponent over periods 25–250 days; β_{G}: median exponent of the GARCH surrogates; GPH: Geweke–Porter-Hudak estimate of β; Low: share of power at periods > 20 days, data and GARCH surrogates. Weekly: number of assets with a significant seven-day (crypto) or five-day (Nifty) component against GARCH after cross-asset correction.

Market	Series	β	R^2	β_{G}	GPH	Low	$\mathrm{Low}_{\mathrm{G}}$	Weekly
Crypto (10)	returns	0.11	0.03	0.00	0.23	0.09	0.09	0/10
	absolute returns	0.45	0.23	1.03	0.86	0.22	0.27	10/10
	squared returns	0.19	0.17	0.74	0.52	0.15	0.20	0/10
	realised vol (20d)	1.60	0.77	2.28	0.97	0.94	0.97	5/10
Nifty (51)	returns	0.03	0.08	0.00	−0.09	0.09	0.09	0/51
	absolute returns	0.70	0.61	0.61	0.64	0.20	0.17	0/51
	squared returns	0.71	0.69	0.52	0.45	0.20	0.15	0/51
	realised vol (20d)	1.88	0.84	1.86	0.75	0.95	0.95	n.a.

has a significant weekly component (smallest nominal p 0.21 for crypto at seven days and 0.021 for stocks at five days, neither significant after correction).

Volatility spectra rise steeply towards low frequencies, as expected for a persistent volatility process (Figure 3 for Bitcoin). Table 2 reports the spectral summaries. For Nifty stocks the spectral exponent of realised volatility (median 1.88) is almost identical to the median exponent of the GARCH surrogates (1.86), and the observed exponent lies inside the 95% surrogate range in 86% of stocks. For cryptocurrencies the exponent of realised volatility (median 1.60) is lower than that of the GARCH surrogates (2.28) and lies below the 95% surrogate range in seven of ten coins (Figure 4); these differences reach nominal significance in seven coins but do not survive cross-asset correction. The fitted GARCH models of cryptocurrencies are close to integrated (for Bitcoin, $\alpha + \beta = 1.000$ over the full sample), so the surrogates are, if anything, *more* persistent than the data. The low-band share tells the same story with stronger evidence: after cross-asset correction, realised volatility of five cryptocurrencies (DOGE, ETH, SOL, TRX, XRP) has *less* power at periods above 20 days than its GARCH surrogates (median share 0.94 against 0.97), whereas squared returns of 18 Nifty series (including the index) have *more* (median 0.20 against 0.15). Volatility of the two markets thus departs from GARCH(1,1) in opposite directions: cryptocurrency volatility is less persistent than an integrated GARCH implies, and equity volatility contains more slow variation than a GARCH(1,1) captures.

5.5 A weekly cycle in cryptocurrency volatility

The only feature that departs from the GARCH null consistently across assets is a peak at a period of seven days in cryptocurrency volatility. After cross-asset correction, the spectral power of absolute returns at seven days exceeds the GARCH surrogate distribution in all ten cryptocurrencies (nominal p between 0.001 and 0.017; cross-asset $q \leq 0.017$), and that of realised volatility in five (ADA, AVAX, BTC, ETH and TRX). Squared returns show the same direction with weaker evidence (nominal $p < 0.05$ in two coins, none significant after correction), as expected for a statistic dominated by a few extreme days. No Nifty series shows a significant five-day component in absolute or squared returns. Because the effect is in volatility and not in returns, and is present in a market that trades every day of the week but absent in one that is closed at weekends, the natural interpretation is a weekly pattern in trading activity

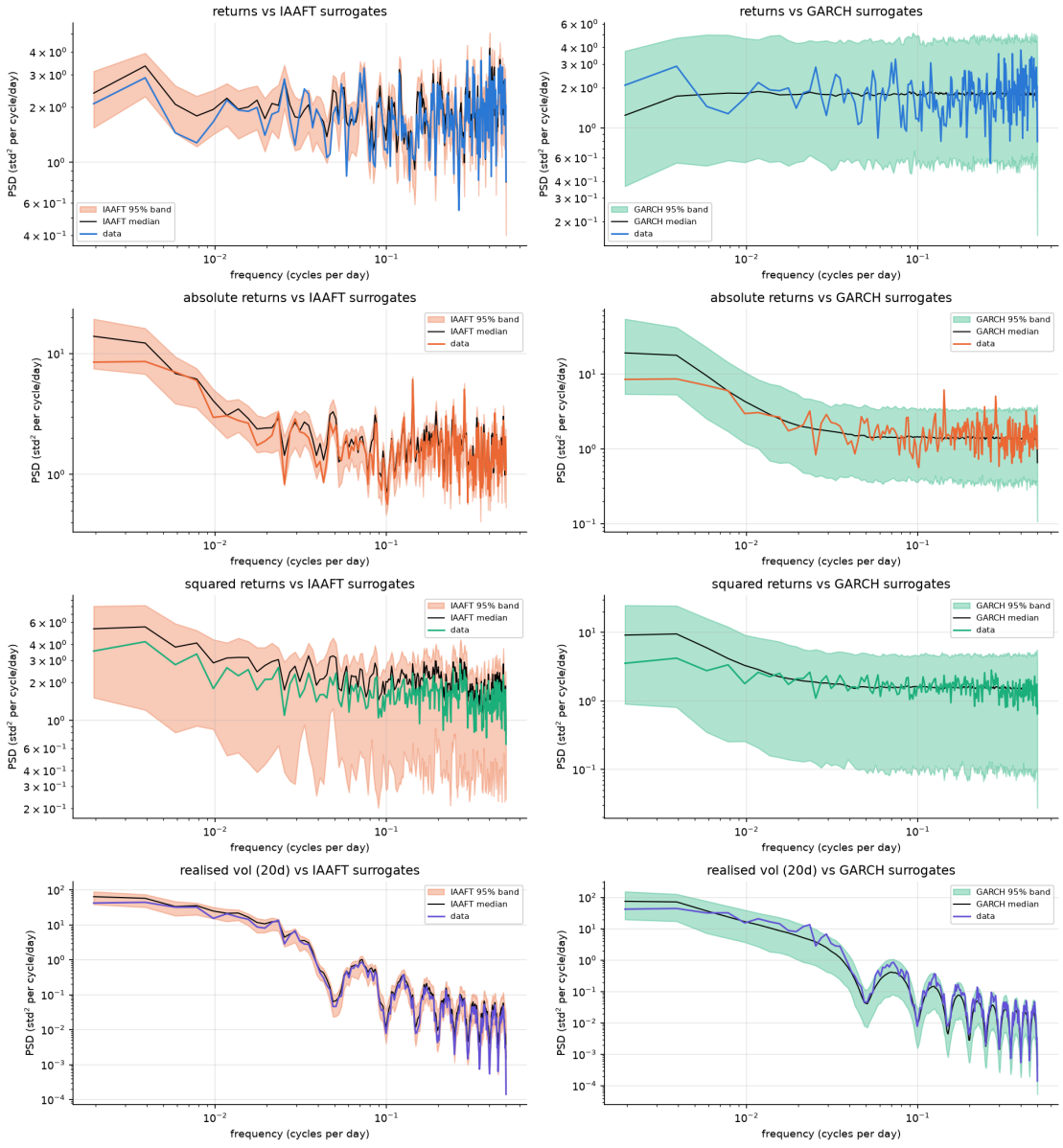


Figure 3: Bitcoin, full sample: Welch spectra of the four series (coloured) with the median (black) and 95% band of 999 IAAFT surrogates (left) and 999 GARCH(1,1)-t surrogates (right). The evenly spaced dips in realised volatility are the zeros of the 20-day moving average at $f = k/20$.

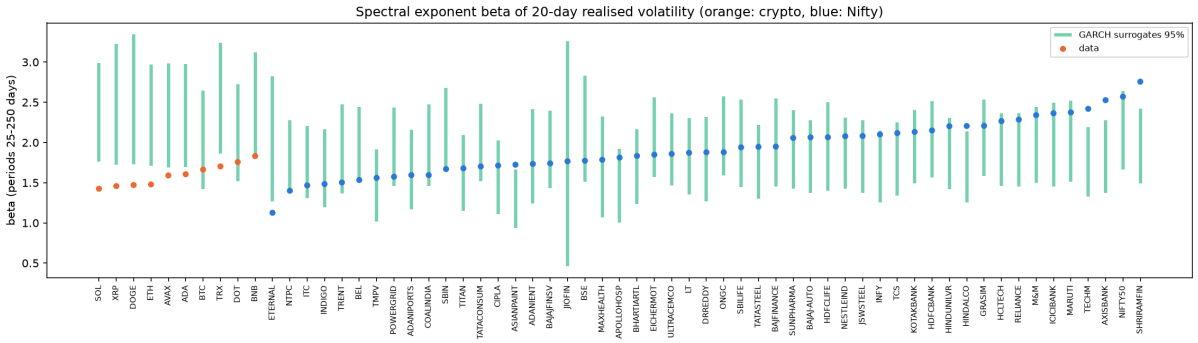


Figure 4: Spectral exponent of 20-day realised volatility for every asset (dots; orange: cryptocurrencies, blue: Nifty) with the 95% range of the exponent over 999 GARCH(1,1)-t surrogates of the same asset (bars).

and volatility of the kind documented by Baur et al. (2019) and Aharon and Qadan (2019). A GARCH(1,1) model without day-of-week terms cannot generate such a pattern, which explains why it is the one feature the null misses. It is a deterministic seasonal component of a stochastic volatility process, not evidence of chaotic dynamics.

5.6 Market periods

Volatility levels differ strongly across periods (Figure 5). The median standard deviation of daily cryptocurrency returns is 0.052, 0.068, 0.042 and 0.041 in P1 to P4, and the median excess kurtosis is 3.2, 13.7, 5.6 and 4.1; for Nifty series the corresponding values are 0.017, 0.023, 0.015 and 0.016, and 3.4, 7.3, 2.6 and 3.5. The second period, which contains the COVID-19 crash, is the most volatile and most heavy-tailed in both markets. The spectral *shape*, however, changes little. The low-band share of realised volatility stays between 0.92 and 0.94 in crypto and between 0.89 and 0.95 in Nifty, and its spectral exponent between 1.60 and 1.92 in crypto; in Nifty it is highest in P2 (2.45) and between 1.51 and 1.64 in the other periods. Against period-specific GARCH surrogates, 38 of 932 global spectral tests have nominal $p < 0.05$, close to the 47 expected by chance; after cross-asset correction two remain significant (absolute returns of HINDUNILVR and of the Nifty 50 index in 2016–2019), and no test on the spectral exponent or the low-band share is significant. Within each period, therefore, the spectra are consistent with a GARCH process refitted to that period: the regimes differ in the level and scale of volatility, not in the type of dynamics.

6 Discussion

6.1 What the evidence does and does not show

The central result is that every conventional indicator of chaos (positive Lyapunov exponent, saturating correlation dimension, $K \approx 1$) is present in these data and equally present in series simulated from a GARCH(1,1)-t model. These indicators therefore carry no information about determinism in daily financial data, which confirms, on a larger and more homogeneous sample, the warnings of Osborne and Provenzale (1989), Ruelle (1990) and Hsieh (1991). A study that

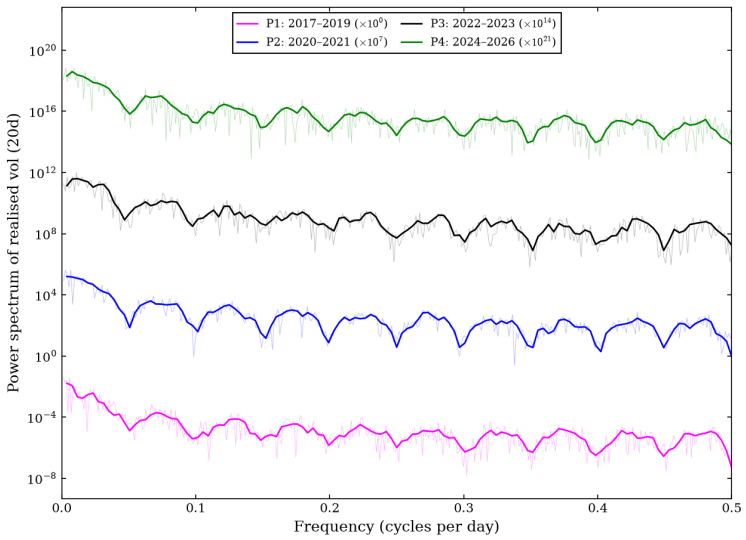


Figure 5: Bitcoin: Welch spectra of 20-day realised volatility in the four market periods, offset vertically by the stated factors for readability (log scale), in the style of Kailasham and Khair (2023).

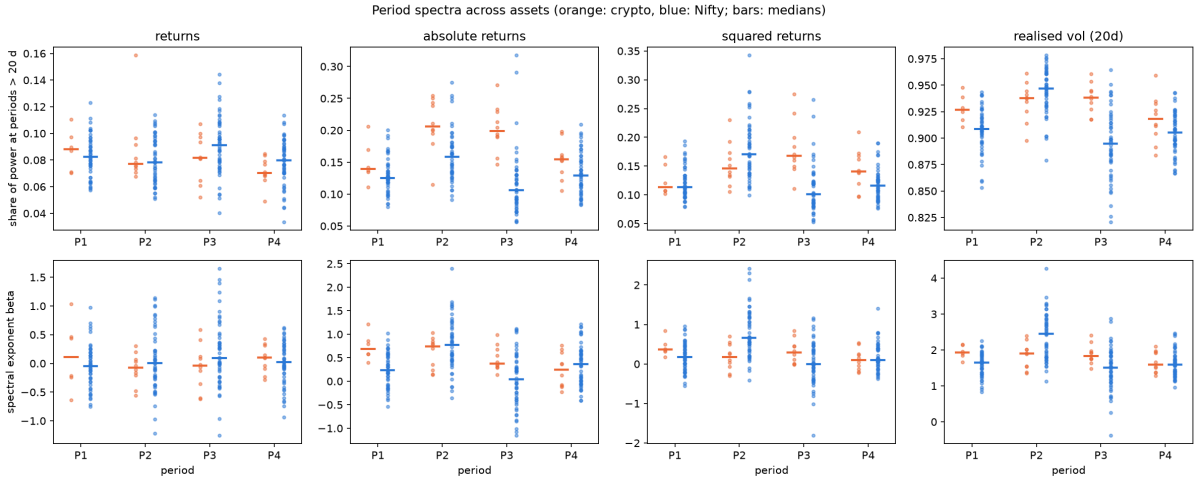


Figure 6: Low-band share (top) and spectral exponent over periods 25–128 days (bottom) in the four market periods for every asset (dots) with market medians (bars); orange: cryptocurrencies, blue: Nifty.

reports chaos in a financial series on the basis of a positive Lyapunov exponent alone, or against a linear or independent null only, is reporting a property of volatility clustering.

The second result is that the null of a linear Gaussian process is rejected widely, and that this rejection is almost entirely accounted for by GARCH. That the Lyapunov exponent of returns is *lower* than in IAAFT surrogates is consistent with this: clustering of large returns makes nearby states in the reconstructed space evolve more similarly than in a linearised, phase-randomised version of the same series. The two significant departures from GARCH after cross-asset correction are both correlation-dimension results, do not co-occur with any predictive statistic and do not replicate across sample halves. We regard them as false positives or as artefacts of misspecification of the GARCH tails, not as findings.

The third result is that the frequency domain is where the GARCH(1,1) null is most informatively wrong. In cryptocurrencies it misses a weekly cycle in volatility and overstates low-frequency persistence; in Indian stocks it understates the low-frequency power of squared

returns in about a third of assets. The latter is consistent with the long-memory and multi-component volatility literature (Ding et al., 1993; Baillie et al., 1996; Corsi, 2009), and with the possibility that slowly changing volatility levels across regimes produce apparent long memory (Lamoureux and Lastrapes, 1990; Mikosch and Stărică, 2004); our period analysis, in which within-period spectra agree with period-specific GARCH fits, supports the second explanation at least in part. The former is consistent with the calendar evidence of Baur et al. (2019) and Aharon and Qadan (2019), here confirmed for ten coins and shown to be the main deficiency of a standard volatility model as a null for nonlinearity tests.

6.2 Interpretation of the two markets

Cryptocurrency and equity volatility differ in more than level. Cryptocurrency volatility shows positive time-reversal asymmetry (increments skewed upwards, that is, sudden increases followed by gradual decreases), a weekly cycle and less persistence than its near-integrated GARCH fit implies. Indian equity volatility shows more slow variation than a GARCH(1,1) captures and, in most of the significant cases, time-reversal asymmetry of the opposite sign, whose mechanism we leave for further work. None of these differences involves deterministic dynamics; all are statements about the structure of a stochastic volatility process. They nevertheless matter for practice: volatility models for cryptocurrencies should include calendar terms, and models for equities may benefit from a slow volatility component.

7 Conclusion

Using a uniform protocol, two surrogate families and false discovery rate control across 61 assets, we find no evidence of low-dimensional deterministic chaos in the daily returns or volatility of ten major cryptocurrencies or of the Nifty 50 stocks and index. The usual indicators of chaos are reproduced by a GARCH(1,1)-t model, the widespread nonlinearity relative to a linear null is explained by conditional heteroskedasticity, and no series meets the criteria for a chaotic classification. The departures from GARCH that do survive are spectral and have a clear economic interpretation: a weekly cycle in cryptocurrency volatility, and market-specific differences in the low-frequency persistence of volatility. Future work should calibrate the power of the protocol on synthetic chaotic systems contaminated by realistic volatility noise, extend the null model to include calendar effects and long memory, and examine intraday data, where the number of observations is large enough to give the diagnostics a fairer chance.

Data and code availability

All results were produced with a single configuration file and fixed random seeds; per-asset tables, surrogate test results and figures are stored with the analysis code. The data were obtained from public sources and were not modified.

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